Integrated Project

Building Multi-Agent AI Systems

Prof. Hakim Hafidi | 4 weeks | Teams of 3

The Project

Design, build, and evaluate a **multi-agent system** where specialized AI agents collaborate to solve a real-world problem. At least one agent must use a **deep learning model you trained** as a functional tool — not decoration. The system must include error handling, logging, and a human-in-the-loop checkpoint.

Technical Stack

Component	Technology	Note
Agent framework	CrewAI (primary)	LangGraph for stretch goal
LLM backend	Gemini free API / Ollama	No personal spending required
DL framework	PyTorch ≥ 2.0	Consistent with course
Language	Python 3.10+	Colab or local

Timeline

	Focus	Deliverables
W1	**Agent Foundations & Scoping**	Working single-agent prototype, architecture doc, domain + dataset chosen
W2	**DL Model as Agent Tool**	Trained model with evaluation, model wrapped as tool, agent uses model in its loop
W3	**Multi-Agent Orchestration**	2+ agents + orchestrator, human-in-the-loop, end-to-end demo
W4	**Evaluation & Defense**	Guardrails, edge-case testing, final report, live demo + 10-min Q&A

Suggested Domains

Choose one (or propose your own with approval). Each naturally decomposes into collaborating agents:

- **A. Smart Document Analyst** — Classify documents (CNN), extract + summarize, generate reports
- **B. Product Review Intelligence** — Sentiment analysis (fine-tuned BERT), competitive search, market brief
- **C. Campus Q&A Assistant** — Intent classification, RAG retrieval, conversation management
- **D. Medical Image Triage** — Image classification (CNN), clinical guideline retrieval, structured report
- **E. Multilingual Content Adapter** — Difficulty classifier, translation + adaptation, quiz generation
- **F. Open Domain** — Your proposal, approved by end of Week 1

Assessment

Component	Weight	Key criteria
System Design & Architecture	**15%**	Agent roles justified, communication diagram, design rationale
DL Model Integration	**15%**	Appropriate model, rigorous evaluation, meaningful integration
Working Multi-Agent System	**20%**	Runs end-to-end, genuine collaboration, code quality, HITL present
Evaluation & Robustness	**10%**	Edge cases tested, honest failure analysis, guardrails, logging
Presentation & Report	**40%**	Live demo works, can defend decisions, 8–12 page report

Minimum Requirements

- **Agents:** 2 specialists + 1 orchestrator, each with distinct role and tools
- **DL model:** PyTorch, trained/fine-tuned by you, with evaluation (accuracy, confusion matrix)
- **Tools:** At least 2 (including your DL model), with clear input/output schemas
- **Human-in-the-loop:** At least 1 checkpoint requiring human approval
- **Error handling:** Tool failures, invalid inputs, unexpected LLM outputs — no crashes
- **Logging:** Every agent action logged with timestamps (JSON)
- **Reproducibility:** README with setup instructions

Submission: GitHub repo + trained model + PDF report (8–12 pages) + demo video (3–5 min) + slides. Final deadline: end of Week 4. Oral defense: 15 min + 10 min Q&A.

Academic integrity: LLM assistants encouraged as tools. You must understand and explain every line of your code during the oral defense.

"The goal is not to build the most complex system. It is to build a system where every component exists for a reason, every agent has a job, and you can explain why."